

TT : 1

Time : 30 minutes

1. Explain sampling distribution and standard error?
2. Establish the relation between Chi-square, F and t distribution?
3. Comparative Chi-square, F and t distribution?

TT 2

Time : 30 minutes

1. Short note on Sampling Distribution?
2. Write a short note on Test of hypothesis?

Shahjalal University of Science and Technology, Sylhet
Department of Statistics
2nd Year 1st Semester 2025(Term Test-II)
Course No. STA 0541 2151
Full Marks: 20, Time: 45 Minutes

1. Using the probability expansion in p and q , derive the pmf of the logarithmic series distribution and show how the geometric and negative binomial distributions follow from the same expansion. 10
2. Derive the probability density function (pdf) of the normal distribution 10

Term Test 01E

Course Code: MAT05412105

Time 40min

Marks: 10

Q1	Define differential equation and integrating factor. Show that the given differential equation $(2x \cos y + 3x^2 y)dx + (x^3 - x^2 \sin y - y)dy = 0, y(0) = 2$ is an exact differential equation and solve it.	4
Q2	Define Bernoulli's differential equation and solve that following differential equation $\frac{dy}{dx} = x^3 y^3 - xy$	6

Term Test 01

Course Code: MAT05412105

Time 30min

Marks: 10

Q1	Define integration factor . Show that the following differential equation is an exact differential equation and also solve it $(x^3 - x^2 \sin y - y)dy + (2x \cos y + 3x^2 y)dx = 0; y(0) = 2$	CO1	5
Q2	Define UC function. Solve the following differential equation $\frac{d^2 y}{dt^2} - 2 \frac{dy}{dt} - 3y = -4e^t - 10 \sin t$	CO2	5

Set A

ECO0311 2107|| Course: Principles of Microeconomics|| Time: 45 minutes|| Marks: 20 (Answer one)

1. (a) Define scarcity and resources. In a world of scarcity, what are the ways to refrain the unlimited wants? $2+2=4$

(b) Define opportunity cost. Explain the relationship between scarcity and opportunity cost. $2+2=4$

(c) How do people make decisions at the margin? Use MC-MB analysis. $2+2=5$

(d) Define productivity possibility frontier (PPF). The following table sets out the North Korea's PPF where Korea produces Missile and Wheat. (i) Draw a graph of North Korea's PPF. Define scarcity and efficiency in the PPF. Calculate the opportunity cost of producing 50 kg to 64 kg of wheat. Also show the effect of threat from the other countries on North Korea's PPF. $1+2+2+1+1=7$

Point	Missile (in thousand)	Wheat (thousand kg)
A	0	70
B	1	64
C	2	50
D	3	40
E	4	20
F	5	0

2. (a) How do you represent the law of demand? Derive the market demand curve from individual demand curves. $1+1=2$

(b) What are the reasons that cause the demand curve goes down as prices go up? 3

(c) Explain the reasons behind movement along the demand curve and shift in the demand curve. 5

(d) Define market equilibrium and also define shortage and surplus. Define price ceiling for any good and its effect on the market. $1+4+1+4=10$

Set B

ECO0311 2107|| Course: Principles of Microeconomics|| Time: 45 minutes|| Marks: 20 (Answer one)

(a) Define opportunity cost. How do people make decisions at the margin? Use MC-MB analysis. $2+3=5$

(b) Explain the effects of any incentive on economy. 3

(c) How does opportunity cost change the behavior of human-beings? 2

(b) Define productivity possibility frontier (PPF). The following table sets out the North Korea's PPF where Korea produces Missile and Wheat. (i) Draw a graph of North Korea's PPF. Define scarcity and efficiency in the PPF. Calculate the opportunity cost of producing 50 kg to 64 kg of wheat. Also show the effect of threat from the other countries on North Korea's PPF. $1+2+2+1+1=7$

(e) What are the ways that shift the PPF outward and produce more? Explain. 3

Point	Missile (in thousand)	Wheat (thousand kg)
A	0	70
B	1	64
C	2	50
D	3	40
E	4	20
F	5	0

2. (a) How do you represent the law of demand? Derive the market demand curve from individual demand curves. $1+3=5$

(b) Explain the reasons behind movement along the demand curve and shift in the demand curve. 5

(c) Suppose the fuel cost of Aeroplane has been increased. As a result, the supply of flights will decrease. If demand for flights remains the same, then explain the new market equilibrium price and quantity of flights. 4

(d) The demand and supply function of chicken is given below:

$$Q_d = 63 - 11P_f + 7P_r + 3P_b + 2Y$$

$$Q_s = 149 + 8P - 9P_e$$

Where, P_f = price of fish = 20 Taka/fish, P_b = price of beef = 11 taka/kg, P_e = price of egg = 5 Taka/piece and Y = income = 480 taka. Solve the equilibrium price (P) and Quantity (Q). Can you identify whether fish or beef are the substitutes/complements of chicken from the demand function? $5+1=6$

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Department of Statistics

2nd Year 1st Semester 2025(Term Test-I)

Course No. STA 0541 2151

Full Marks: 20, Time: 40 Minutes

1. Explain the concepts of population, census data, and sample space with examples. 4
2. Explain how the probability density function (pdf) of the exponential distribution is derived starting from the Poisson distribution. 8
3. Derive the probability mass function (pmf) of the negative binomial distribution and explain the origin of the name “negative binomial.” 8

Term Test 02

Course Code: MAT05412105

Time 30min

Marks: 10

Q1	Define Laplace transform. Find the Laplace transform of the function $(i)\mathcal{L}\{(5e^{2t} - 3)^2\}$ $(ii)\mathcal{L}\{t^2 \sin t\}$	5
	OR	
	Define the gamma functions. Evaluate $\int_0^\infty t^{\frac{-3}{2}}(1 - e^{-t})dt$	5
Q2	Use Convolution theorem to find $\mathcal{L}^{-1}\left\{\frac{1}{(s+1)(s^2+1)}\right\}$	5
	OR	
	Expand $f(x) = \begin{cases} 0 & ; -\pi < x < 0 \\ \pi - x & ; 0 \leq x < \pi \end{cases}$ in a Fourier series	5

Set A

ECO0311 2107|| Course: Principles of Microeconomics|| Time: 45 minutes|| Marks: 15

1. (a) Define price elasticity, income elasticity and cross elasticity of demand with appropriate formula. 4
- (b) Explain the relationship between elasticity and the slope of the demand function from the following demand function: $Q_d = a - bp$. Consider the case of point elasticity. 3
- (c) Explain the determinants of price elasticity of demand. 4
- (d) Explain the relationship between elasticity of demand and the effect of taxation. In which case producers pay the full tax? 4

Set B

ECO0311 2107|| Course: Principles of Microeconomics|| Time: 45 minutes|| Marks: 15

1. (a) Define price elasticity, income elasticity and cross elasticity of demand with appropriate formula. 4
- (b) Why the elasticity is different along the linear demand curve---explain using the concept of price elasticity of demand. 4
- (c) Consider the following information: $P_1 = 10$, $Q_1 = 200$, $P_2 = 30$, $Q_2 = 180$, $Y_1 = 300$, $Y_2 = 400$. Calculate the price elasticity and income elasticity of demand. Explain what type of good it is in terms of price and income. 4
- (d) Consider that the price of 'tobacco leaves' increase. Can it reduce the demand for 'tobacco leaves'? Using the concept of elasticity and graph, explain. 3

Shahjalal University of Science and Technology, Sylhet
Department of Statistics
2nd Year 1st Semester 2025(Term Test-III)
Course No. STA 0541 2151
Full Marks: 20, Time: 35 Minutes

1. Explain why the Lognormal and Pareto distributions are used in modeling income distribution, discussing the limitations of the Normal distribution and justifying the preference for the geometric mean over the arithmetic mean in skewed and heavy-tailed income data. 8
2. Find the mean, variance and geometric mean of Pareto distribution. 12